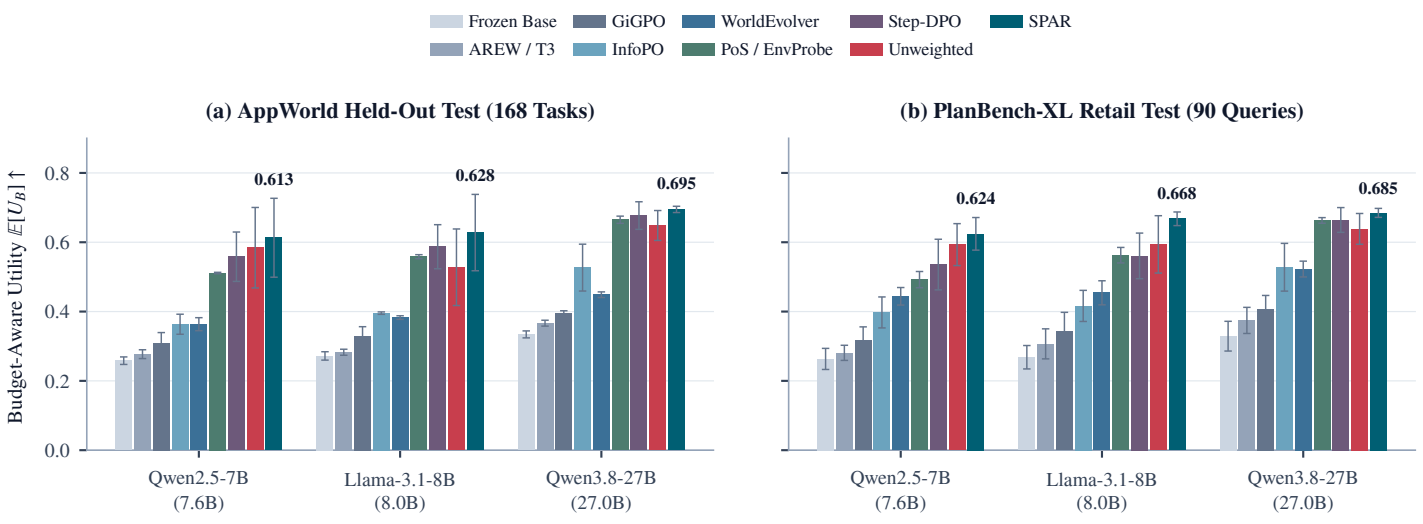
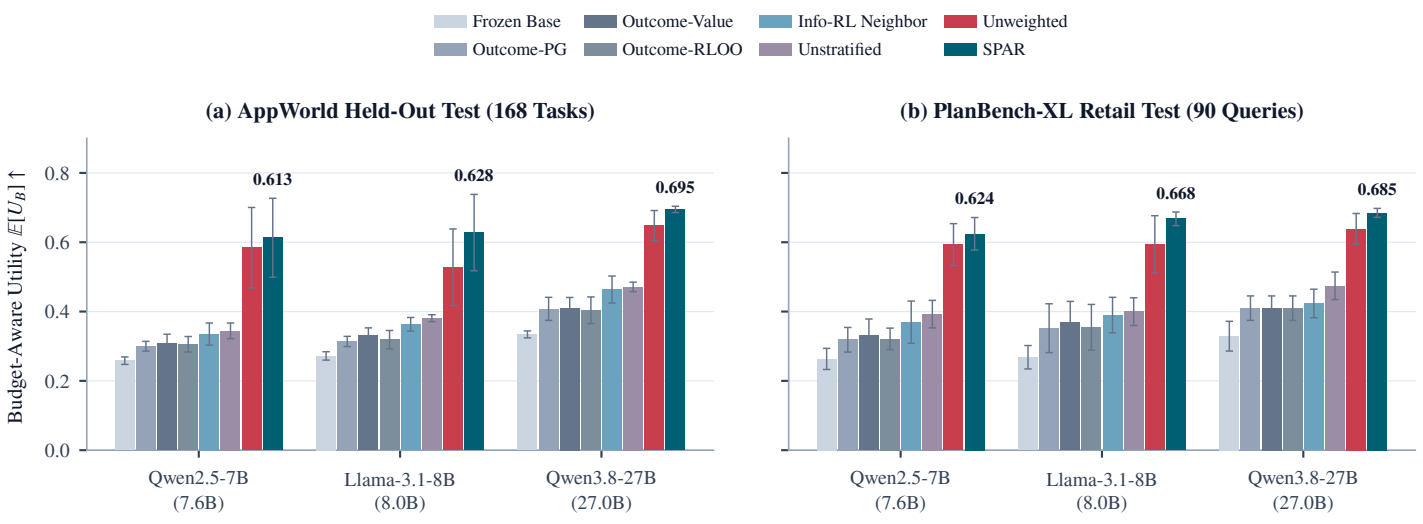
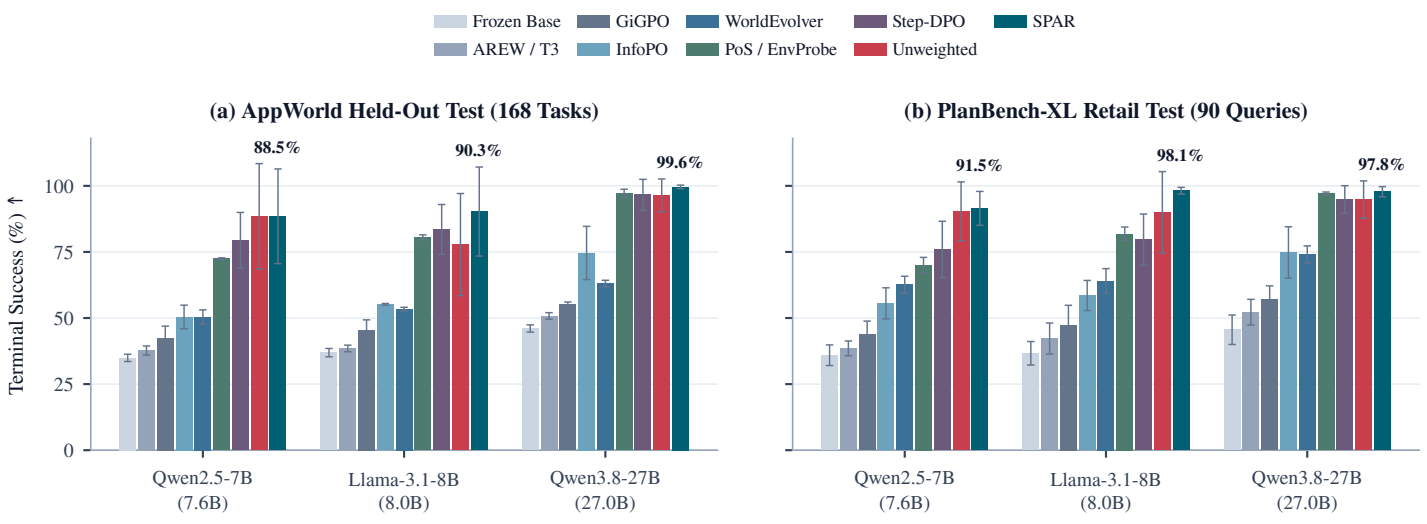
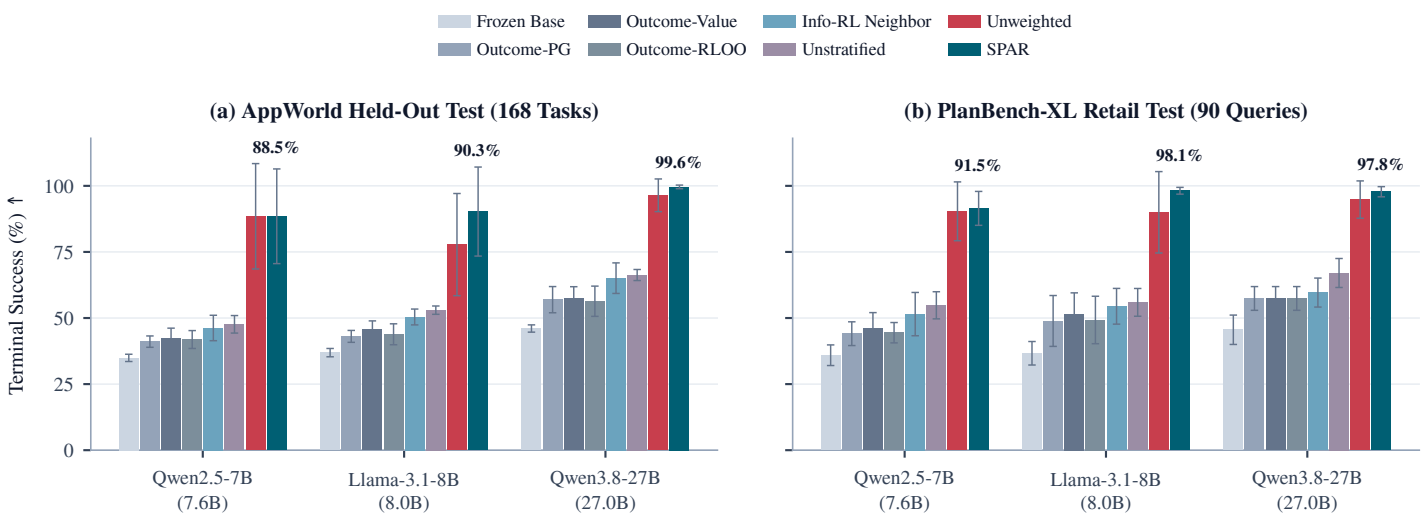
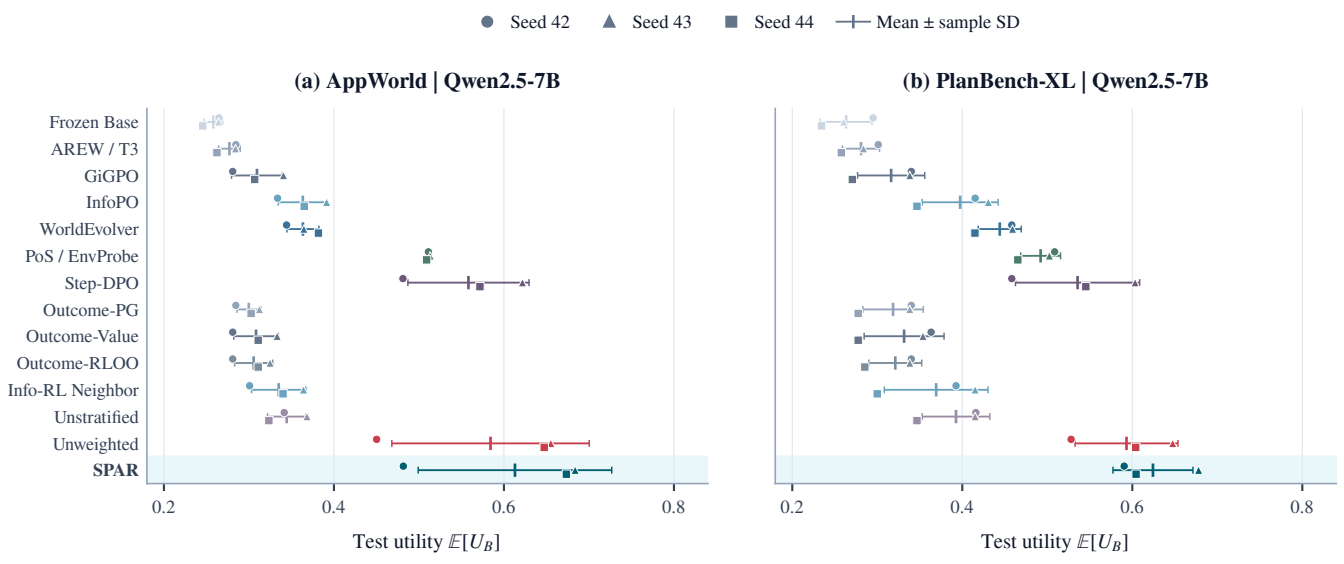
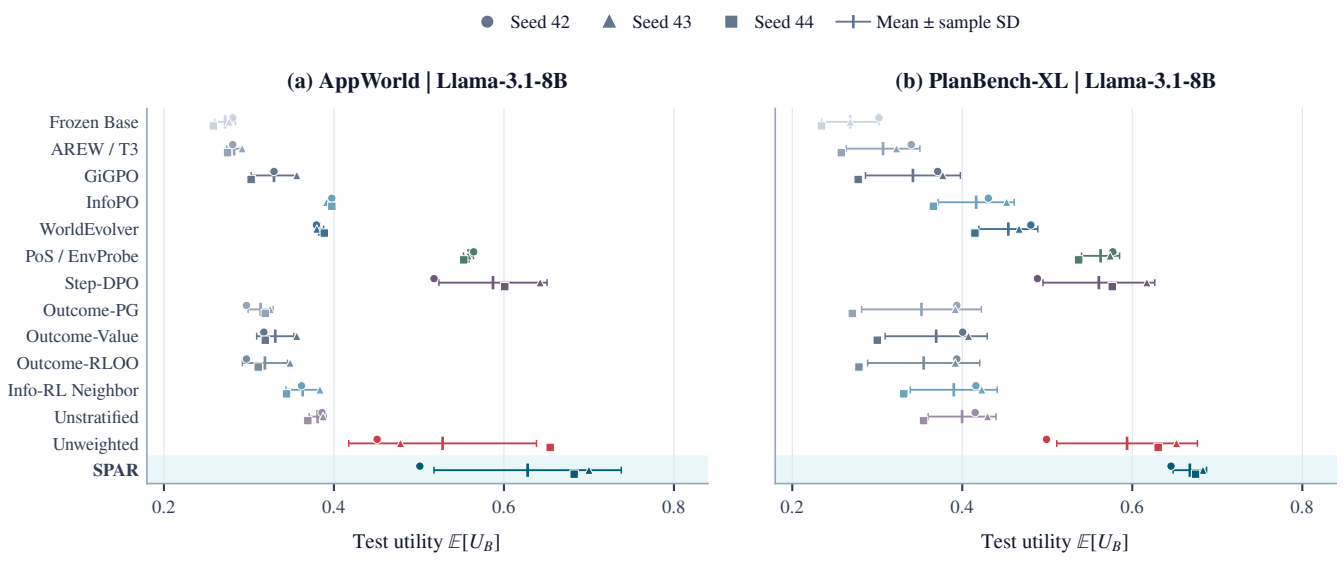

Frozen Base
GiGPO
WorldEvolver
Step-DPO
SPAR
AREW / T3
InfoPO
PoS / EnvProbe
Unweighted
(a) AppWorld Held-Out Test (168 Tasks)
(b) PlanBench-XL Retail Test (90 Queries)
Budget-Aware Utility $\mathbb{E}[U_B] \uparrow$
0.0
0.2
0.4
0.6
0.8
0.613
0.628
0.695
0.624
0.668
0.685
Qwen2.5-7B
(7.6B)
Llama-3.1-8B
(8.0B)
Qwen3.8-27B
(27.0B)

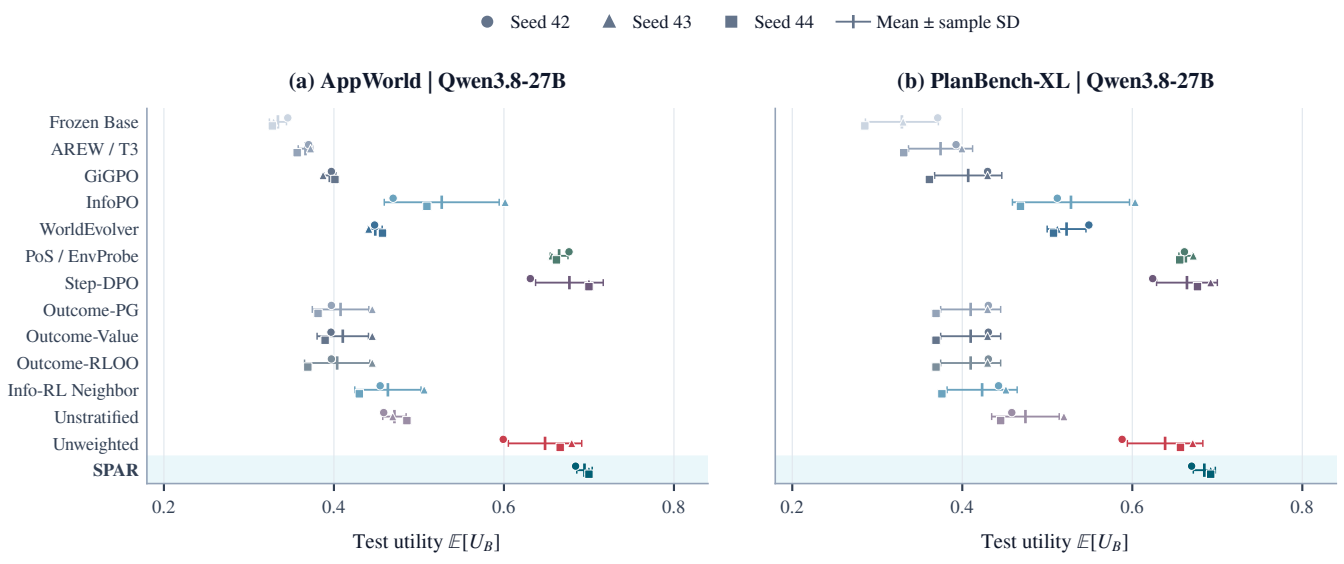

Frozen Base
Outcome-Value
Info-RL Neighbor
Unweighted
Outcome-PG
Outcome-RLOO
Unstratified
SPAR
(a) AppWorld Held-Out Test (168 Tasks)
(b) PlanBench-XL Retail Test (90 Queries)
Budget-Aware Utility $\mathbb{E}[U_B] \uparrow$
0.0
0.2
0.4
0.6
0.8
0.613
0.628
0.695
0.624
0.668
0.685
Qwen2.5-7B
(7.6B)
Llama-3.1-8B
(8.0B)
Qwen3.8-27B
(27.0B)

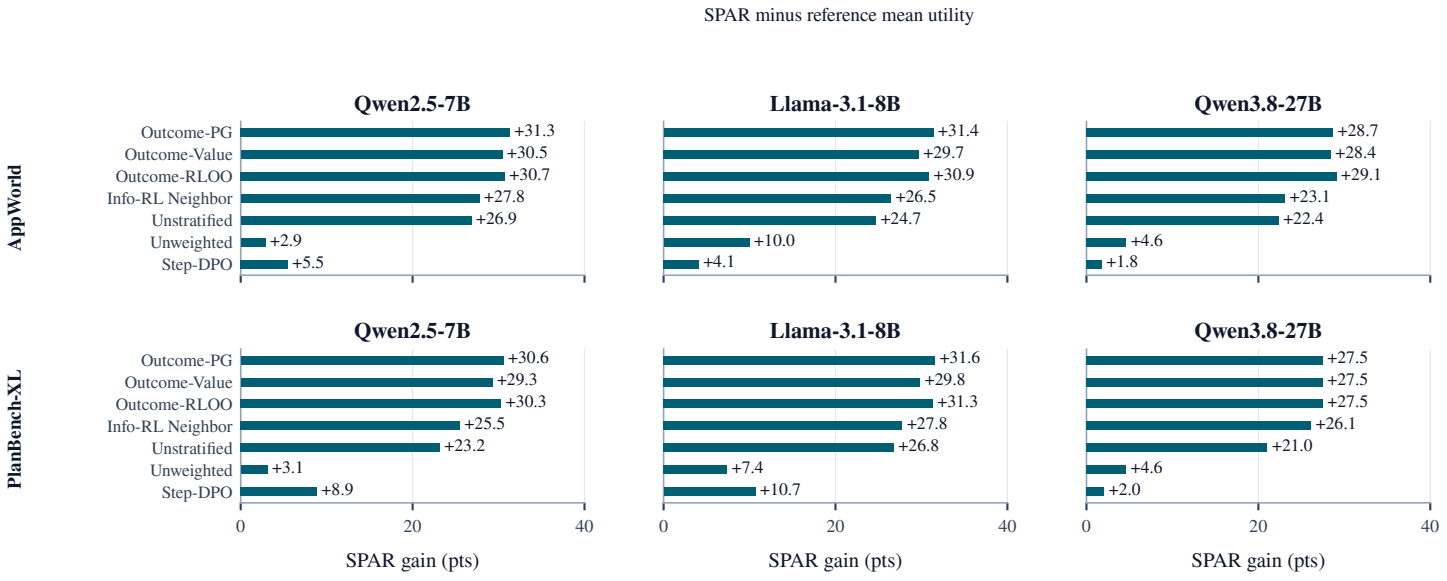

Frozen Base
AREW / T3
GiGPO
InfoPO
WorldEvolver
PoS / EnvProbe
Step-DPO
Unweighted
SPAR
(a) AppWorld Held-Out Test (168 Tasks)
Terminal Success (%) ↑
0
25
50
75
100
88.5%
90.3%
99.6%
Qwen2.5-7B
(7.6B)
Llama-3.1-8B
(8.0B)
Qwen3.8-27B
(27.0B)
(b) PlanBench-XL Retail Test (90 Queries)
91.5%
98.1%
97.8%
Qwen2.5-7B
(7.6B)
Llama-3.1-8B
(8.0B)
Qwen3.8-27B
(27.0B)

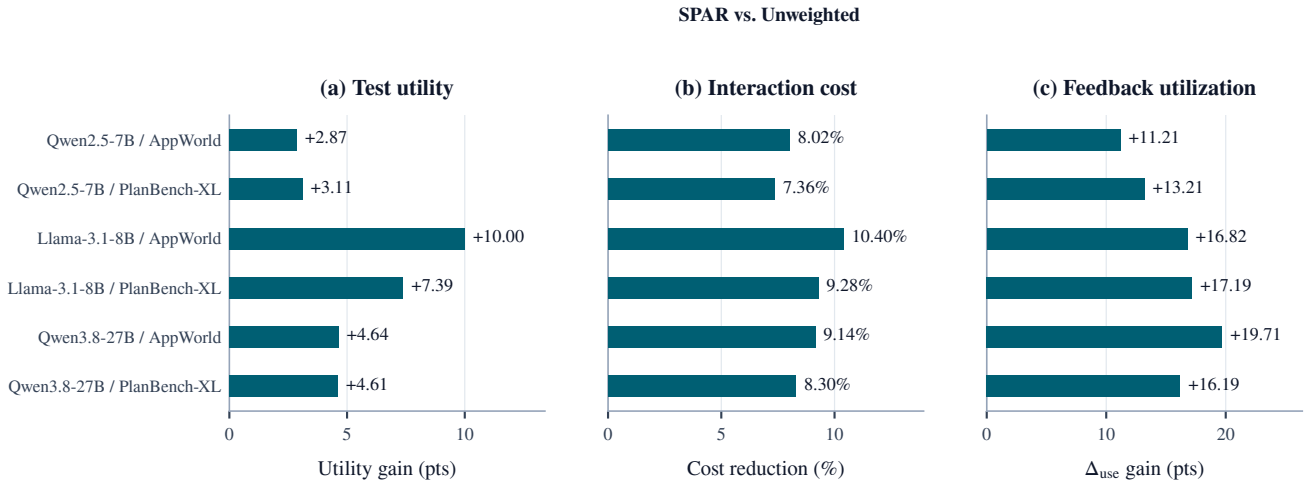

Frozen Base
Outcome-Value
Info-RL Neighbor
Unweighted
Outcome-PG
Outcome-RLOO
Unstratified
SPAR
(a) AppWorld Held-Out Test (168 Tasks)
(b) PlanBench-XL Retail Test (90 Queries)
Terminal Success (%) ↑
100
75
50
25
0
88.5%
90.3%
99.6%
91.5%
98.1%
97.8%
Qwen2.5-7B
(7.6B)
Llama-3.1-8B
(8.0B)
Qwen3.8-27B
(27.0B)

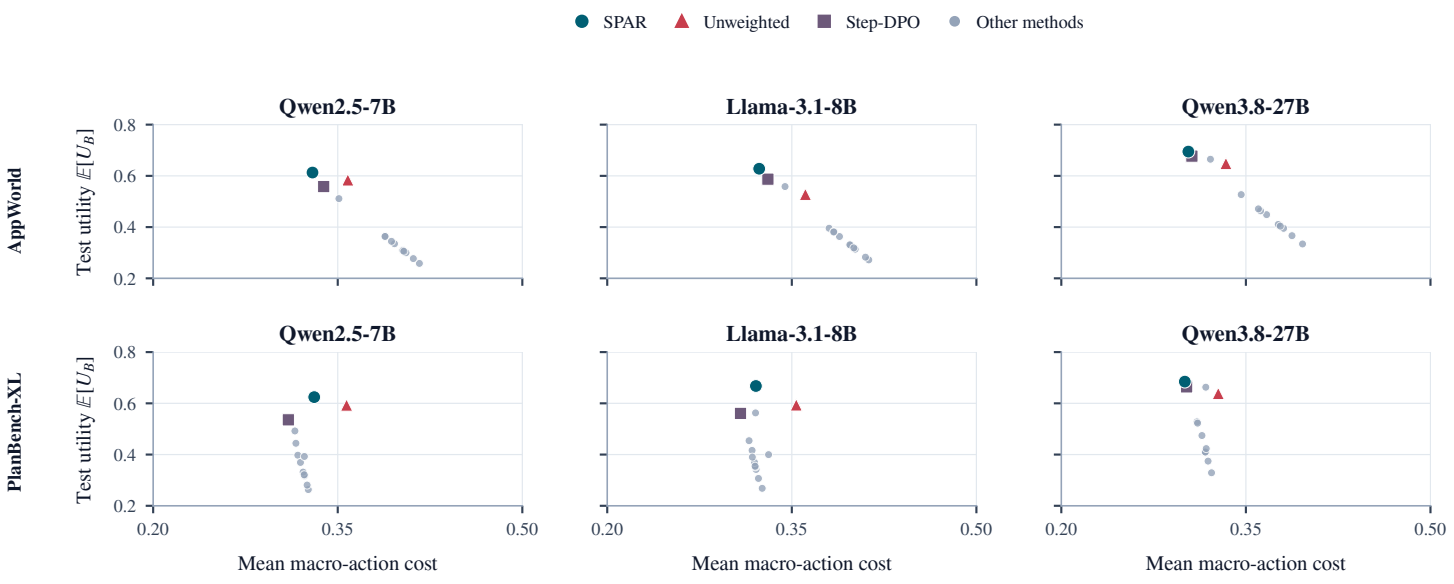

Seed 42
Seed 43
Seed 44
Mean ± sample SD
(a) AppWorld | Qwen2.5-7B
(b) PlanBench-XL | Qwen2.5-7B
Frozen Base
AREW / T3
GiGPO
InfoPO
WorldEvolver
PoS / EnvProbe
Step-DPO
Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
SPAR
0.2
0.4
0.6
0.8
Test utility $\mathbb{E}[U_B]$

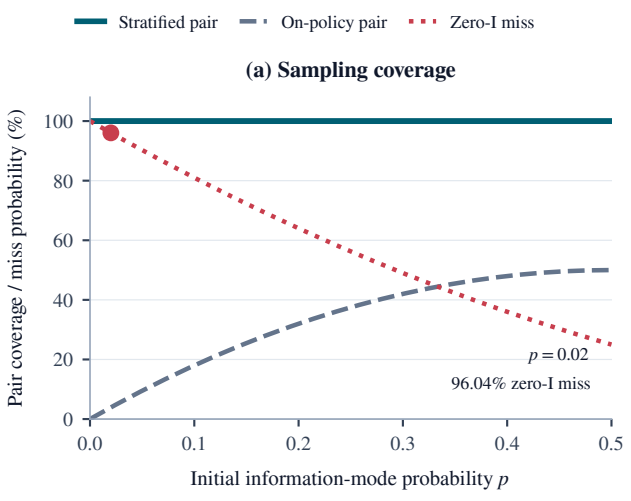

Seed 42
Seed 43
Seed 44
Mean ± sample SD
(a) AppWorld | Llama-3.1-8B
(b) PlanBench-XL | Llama-3.1-8B
Frozen Base
AREW / T3
GiGPO
InfoPO
WorldEvolver
PoS / EnvProbe
Step-DPO
Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
SPAR
0.2
0.4
0.6
0.8
Test utility $\mathbb{E}[U_B]$


Seed 42
Seed 43
Seed 44
Mean ± sample SD
(a) AppWorld | Qwen3.8-27B
(b) PlanBench-XL | Qwen3.8-27B
Frozen Base
AREW / T3
GiGPO
InfoPO
WorldEvolver
PoS / EnvProbe
Step-DPO
Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
SPAR
0.2
0.4
0.6
0.8
Test utility $\mathbb{E}[U_B]$


SPAR minus reference mean utility
AppWorld
PlanBench-XL
Qwen2.5-7B
Llama-3.1-8B
Qwen3.8-27B
Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
Step-DPO
+31.3
+30.5
+30.7
+27.8
+26.9
+2.9
+5.5
+31.4
+29.7
+30.9
+26.5
+24.7
+10.0
+4.1
+28.7
+28.4
+29.1
+23.1
+22.4
+4.6
+1.8
+30.6
+29.3
+30.3
+25.5
+23.2
+3.1
+8.9
+31.6
+29.8
+31.3
+27.8
+26.8
+7.4
+10.7
+27.5
+27.5
+27.5
+26.1
+21.0
+4.6
+2.0
0
20
40
SPAR gain (pts)


SPAR vs. Unweighted
(a) Test utility
(b) Interaction cost
(c) Feedback utilization
Qwen2.5-7B / AppWorld
Qwen2.5-7B / PlanBench-XL
Llama-3.1-8B / AppWorld
Llama-3.1-8B / PlanBench-XL
Qwen3.8-27B / AppWorld
Qwen3.8-27B / PlanBench-XL
+2.87
+3.11
+10.00
+7.39
+4.64
+4.61
8.02%
7.36%
10.40%
9.28%
9.14%
8.30%
+11.21
+13.21
+16.82
+17.19
+19.71
+16.19
0
5
10
0
5
10
0
10
20
Utility gain (pts)
Cost reduction (%)
Δuse gain (pts)


SPAR
Unweighted
Step-DPO
Other methods
AppWorld
PlanBench-XL
Qwen2.5-7B
Llama-3.1-8B
Qwen3.8-27B
Test utility $\mathbb{E}[U_B]$
0.8
0.6
0.4
0.2
0.20
0.35
0.50
Mean macro-action cost

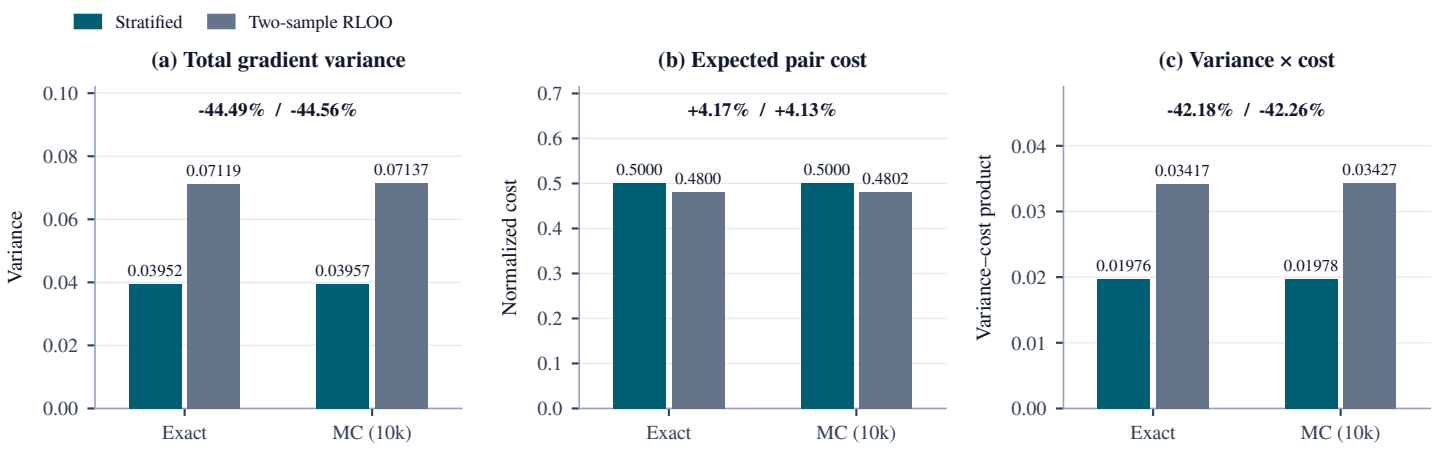


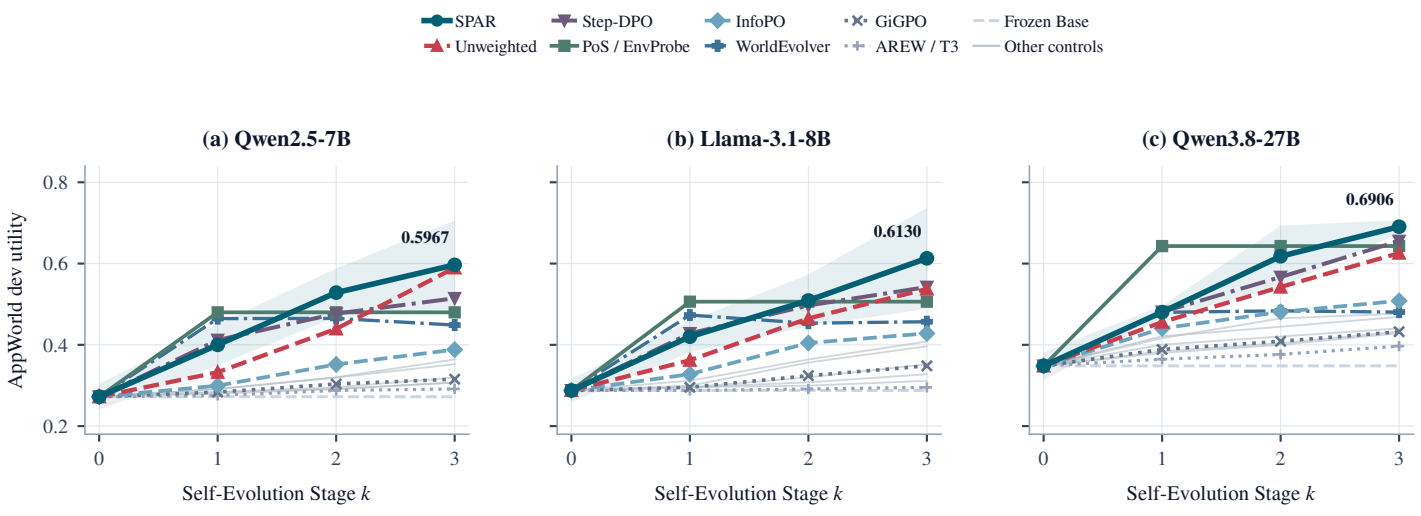
(b) Gradients (MC: 10,000)

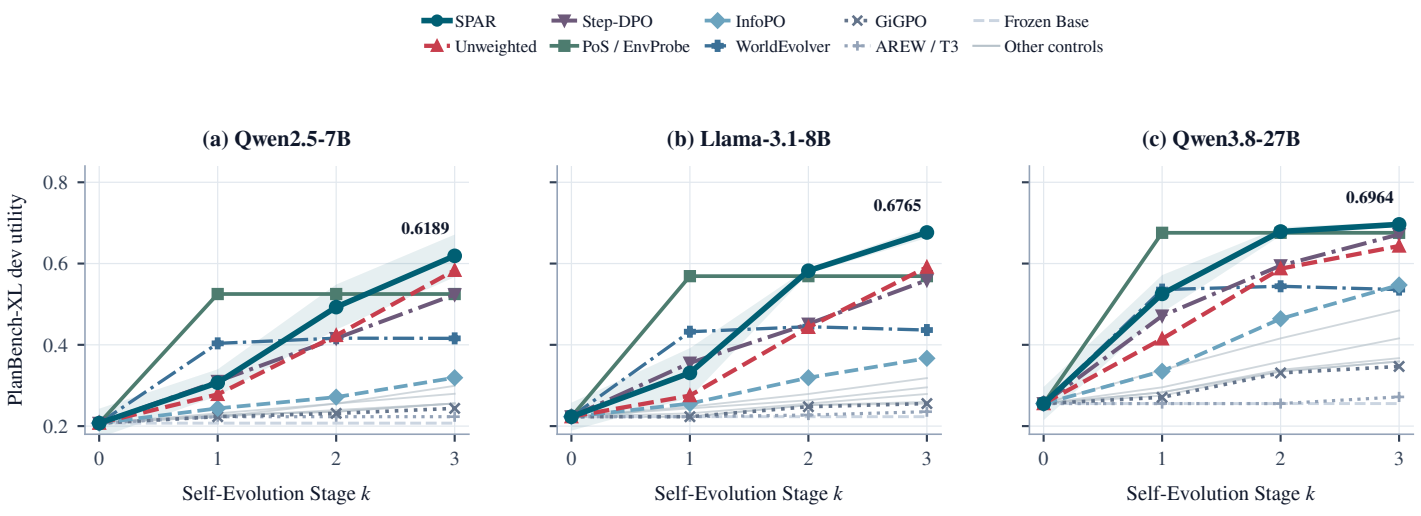
Exact objective: 58/125 = 0.4640

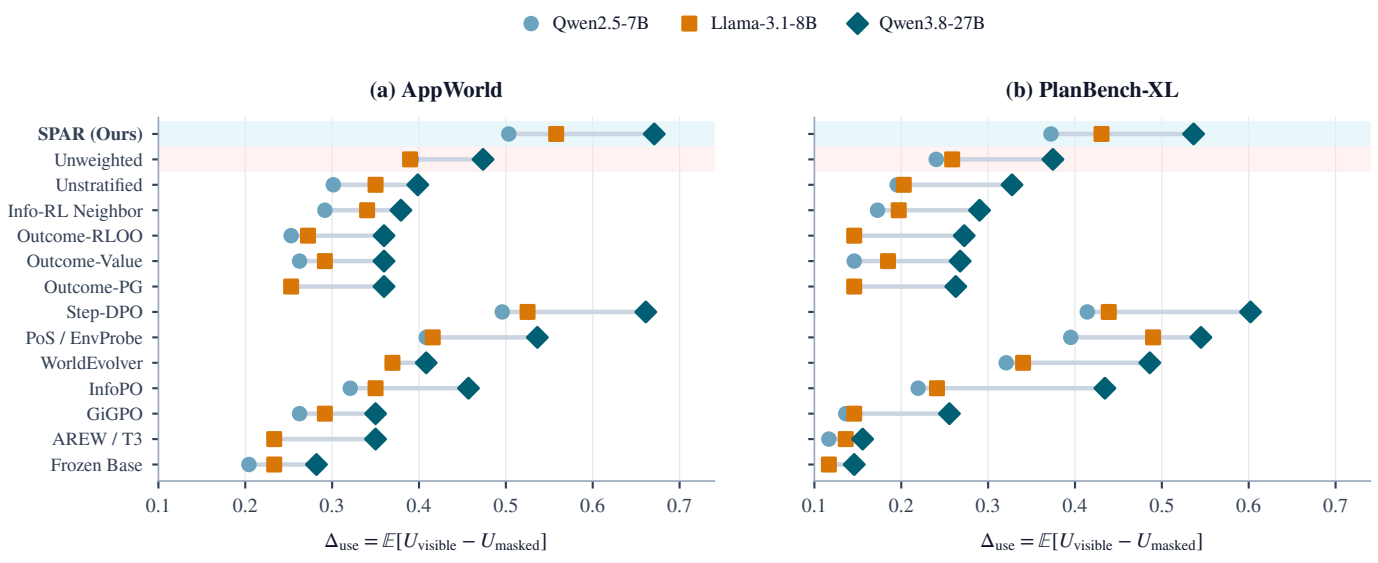
Estimator	g_1	g_2	g_3
Analytical	+0.03840	+0.04480	+0.00000
Stratified exact	+0.03840	+0.04480	+0.00000
RLOO exact	+0.03840	+0.04480	+0.00000
Autograd	+0.03840	+0.04480	+0.00000
Stratified MC	+0.03814	+0.04439	-0.00031
RLOO MC	+0.03736	+0.04529	-0.00164

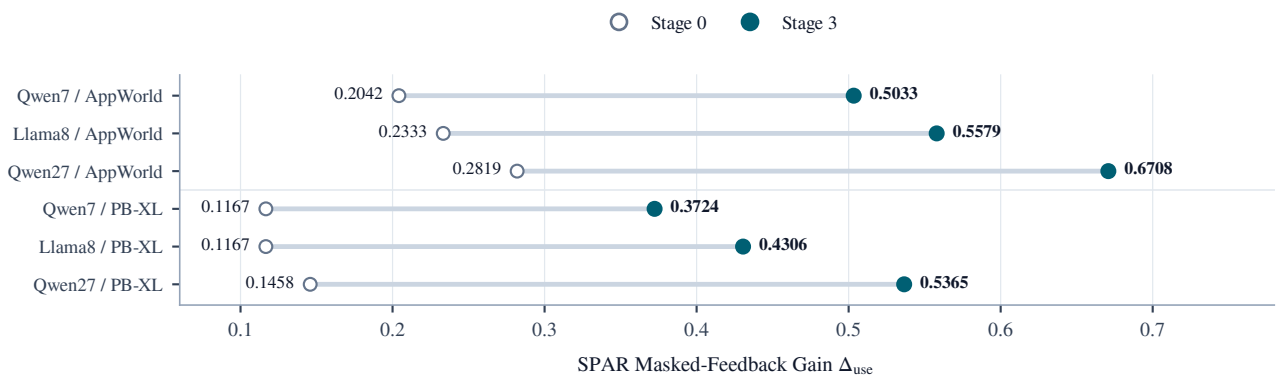
Variance identity: LHS = RHS = 0.0164736

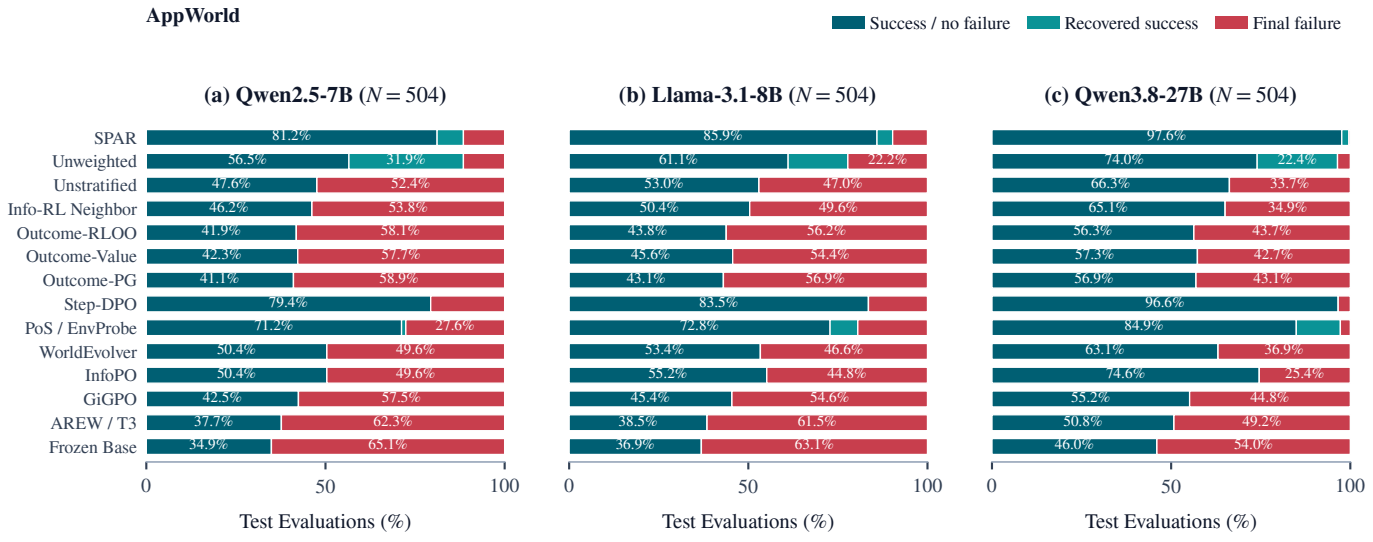

Stratified
Two-sample RLOO
(a) Total gradient variance
-44.49% / -44.56%
Variance
0.10
0.08
0.06
0.04
0.02
0.00
0.07119
0.07137
0.03952
0.03957
Exact
MC (10k)
(b) Expected pair cost
+4.17% / +4.13%
Normalized cost
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.5000
0.4800
0.5000
0.4802
Exact
MC (10k)
(c) Variance × cost
-42.18% / -42.26%
Variance–cost product
0.04
0.03
0.02
0.01
0.00
0.03417
0.03427
0.01976
0.01978
Exact
MC (10k)

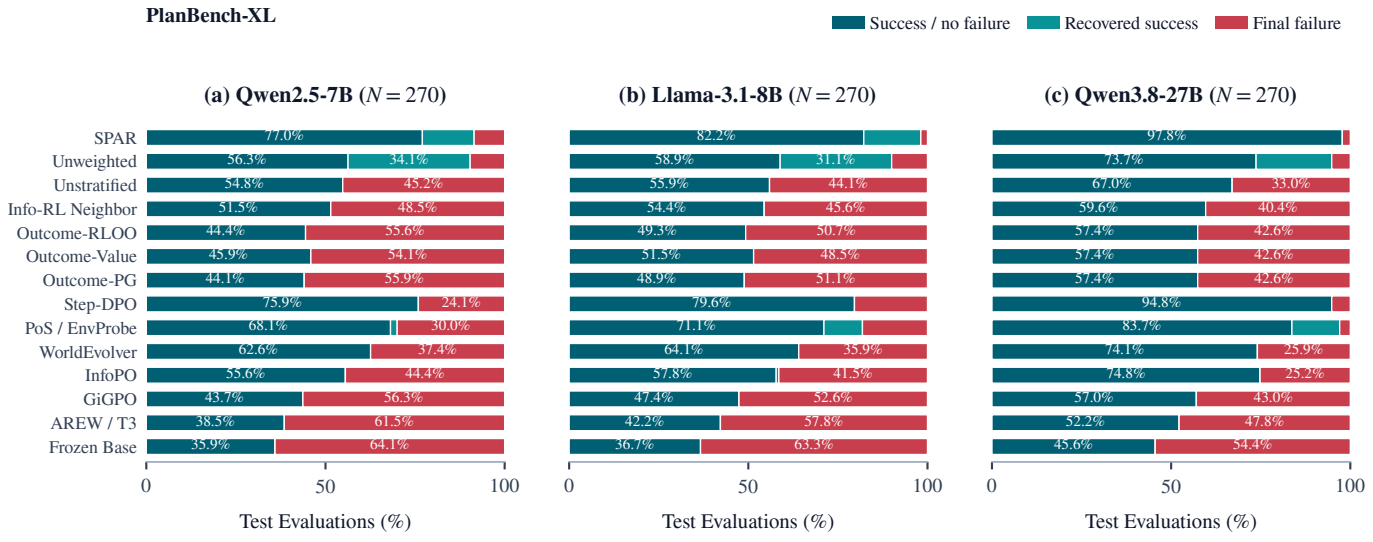

SPAR
Unweighted
Step-DPO
PoS / EnvProbe
InfoPO
WorldEvolver
GiGPO
AREW / T3
Frozen Base
Other controls
(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
AppWorld dev utility
0.8
0.6
0.4
0.2
0.5967
0.6130
0.6906
0
1
2
3
Self-Evolution Stage k


SPAR
Unweighted
Step-DPO
PoS / EnvProbe
InfoPO
WorldEvolver
GiGPO
AREW / T3
Frozen Base
Other controls
(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
PlanBench-XL dev utility
0.8
0.6
0.4
0.2
0.6189
0.6765
0.6964
0
1
2
3
Self-Evolution Stage k


Qwen2.5-7B
Llama-3.1-8B
Qwen3.8-27B
(a) AppWorld
(b) PlanBench-XL
SPAR (Ours)
Unweighted
Unstratified
Info-RL Neighbor
Outcome-RLOO
Outcome-Value
Outcome-PG
Step-DPO
PoS / EnvProbe
WorldEvolver
InfoPO
GiGPO
AREW / T3
Frozen Base
0.1
0.2
0.3
0.4
0.5
0.6
0.7
$\Delta_{\text{use}} = \mathbb{E}[U_{\text{visible}} - U_{\text{masked}}]$

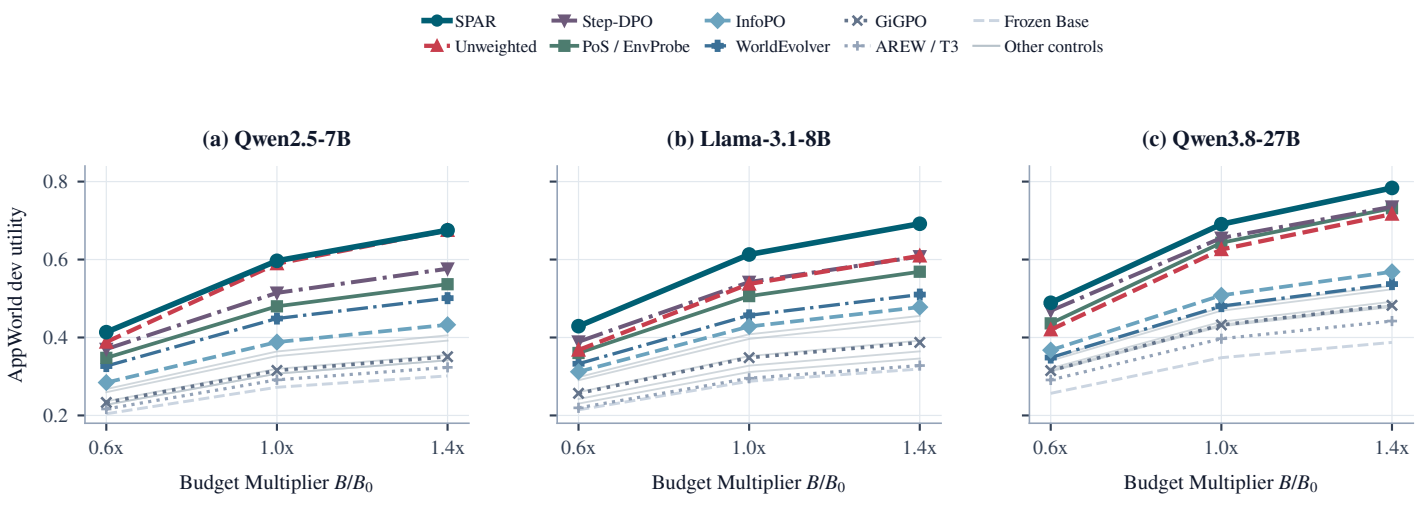

Stage 0
Stage 3
Qwen7 / AppWorld
Llama8 / AppWorld
Qwen27 / AppWorld
Qwen7 / PB-XL
Llama8 / PB-XL
Qwen27 / PB-XL
0.2042
0.5033
0.2333
0.5579
0.2819
0.6708
0.1167
0.3724
0.1167
0.4306
0.1458
0.5365
0.1
0.2
0.3
0.4
0.5
0.6
0.7
SPAR Masked-Feedback Gain Δ_{use}

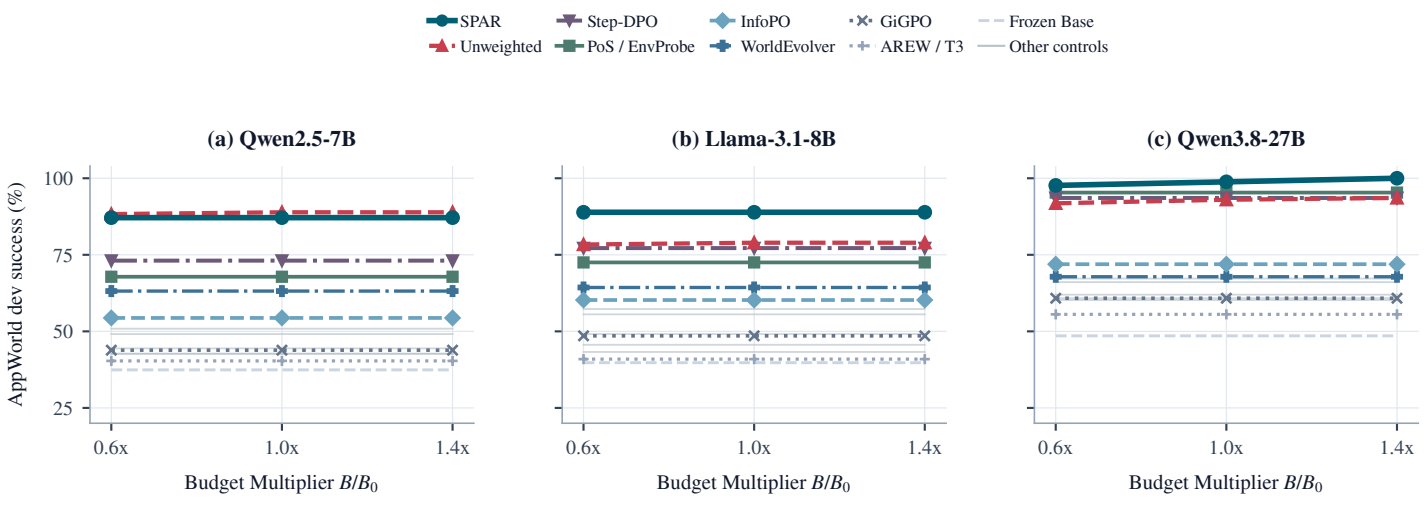

AppWorld
Success / no failure
Recovered success
Final failure
(a) Qwen2.5-7B (N = 504)
(b) Llama-3.1-8B (N = 504)
(c) Qwen3.8-27B (N = 504)
SPAR
Unweighted
Unstratified
Info-RL Neighbor
Outcome-RLOO
Outcome-Value
Outcome-PG
Step-DPO
PoS / EnvProbe
WorldEvolver
InfoPO
GiGPO
AREW / T3
Frozen Base
81.2%
56.5%
31.9%
47.6%
52.4%
46.2%
53.8%
41.9%
58.1%
42.3%
57.7%
41.1%
58.9%
79.4%
71.2%
27.6%
50.4%
49.6%
50.4%
49.6%
42.5%
57.5%
37.7%
62.3%
34.9%
65.1%
85.9%
61.1%
22.2%
53.0%
47.0%
50.4%
49.6%
43.8%
56.2%
45.6%
54.4%
43.1%
56.9%
83.5%
72.8%
53.4%
46.6%
55.2%
44.8%
45.4%
54.6%
38.5%
61.5%
36.9%
63.1%
97.6%
74.0%
22.4%
66.3%
33.7%
65.1%
34.9%
56.3%
43.7%
57.3%
42.7%
56.9%
43.1%
96.6%
84.9%
63.1%
36.9%
74.6%
25.4%
55.2%
44.8%
50.8%
49.2%
46.0%
54.0%
0
50
100
Test Evaluations (%)


PlanBench-XL
Success / no failure
Recovered success
Final failure
(a) Qwen2.5-7B (N = 270)
(b) Llama-3.1-8B (N = 270)
(c) Qwen3.8-27B (N = 270)
SPAR
Unweighted
Unstratified
Info-RL Neighbor
Outcome-RLOO
Outcome-Value
Outcome-PG
Step-DPO
PoS / EnvProbe
WorldEvolver
InfoPO
GiGPO
AREW / T3
Frozen Base
77.0%
56.3%
34.1%
54.8%
45.2%
51.5%
48.5%
44.4%
55.6%
45.9%
54.1%
44.1%
55.9%
75.9%
24.1%
68.1%
30.0%
62.6%
37.4%
55.6%
44.4%
43.7%
56.3%
38.5%
61.5%
35.9%
64.1%
82.2%
58.9%
31.1%
55.9%
44.1%
54.4%
45.6%
49.3%
50.7%
51.5%
48.5%
48.9%
51.1%
79.6%
71.1%
64.1%
35.9%
57.8%
41.5%
47.4%
52.6%
42.2%
57.8%
36.7%
63.3%
97.8%
73.7%
67.0%
33.0%
59.6%
40.4%
57.4%
42.6%
57.4%
42.6%
57.4%
42.6%
94.8%
83.7%
74.1%
25.9%
74.8%
25.2%
57.0%
43.0%
52.2%
47.8%
45.6%
54.4%
0
50
100
Test Evaluations (%)

Recovered / failure cases

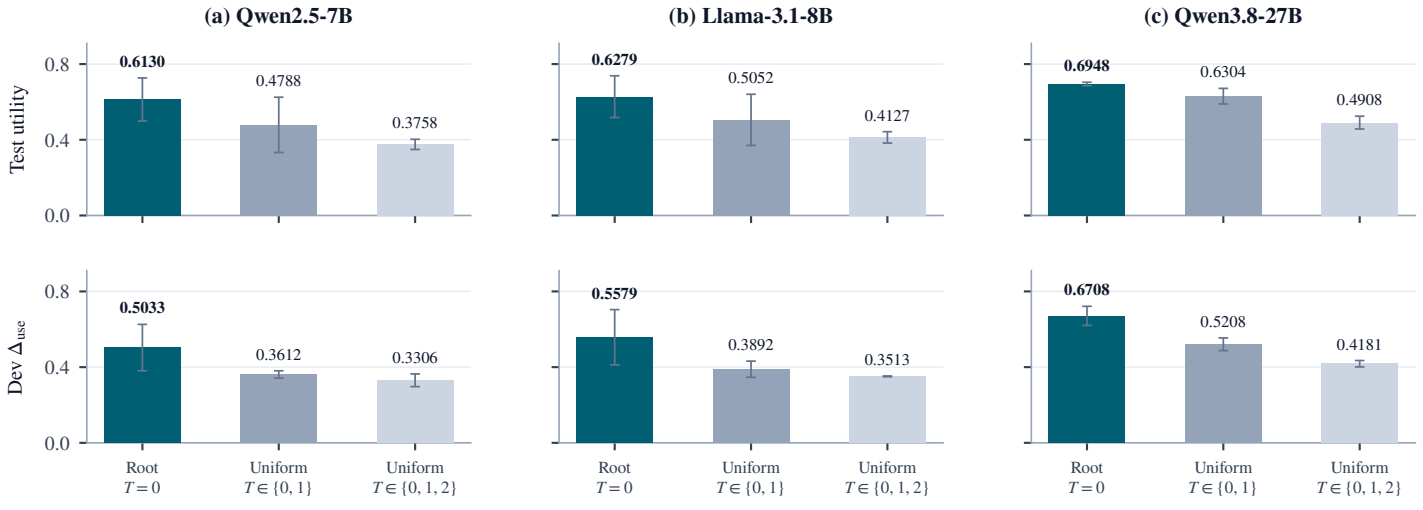
Method	AppWorld			PlanBench-XL		
	Qwen7	Llama8	Qwen27	Qwen7	Llama8	Qwen27
Frozen Base	0/328	0/318	0/272	0/173	0/171	0/147
AREW / T3	0/314	0/310	0/248	0/166	0/156	0/129
GiGPO	0/290	0/275	0/226	0/152	0/142	0/116
InfoPO	0/250	0/226	0/128	0/120	2/114	0/68
WorldEvolver	0/250	0/235	0/186	0/101	0/97	0/70
PoS / EnvProbe	6/145	39/137	62/76	5/86	29/78	36/44
Step-DPO	0/104	0/83	0/17	0/65	0/55	0/14
Outcome-PG	0/297	0/287	0/217	0/151	0/138	0/115
Outcome-Value	0/291	0/274	0/215	0/146	0/131	0/115
Outcome-RLOO	0/293	0/283	0/220	0/150	0/137	0/115
Info-RL Neighbor	0/271	0/250	0/176	0/131	0/123	0/109
Unstratified	0/264	0/237	0/170	0/122	0/119	0/89
Unweighted	161/219	84/196	113/131	92/118	84/111	57/71
SPAR	**37/95**	**22/71**	**10/12**	**39/62**	**43/48**	**0/6**

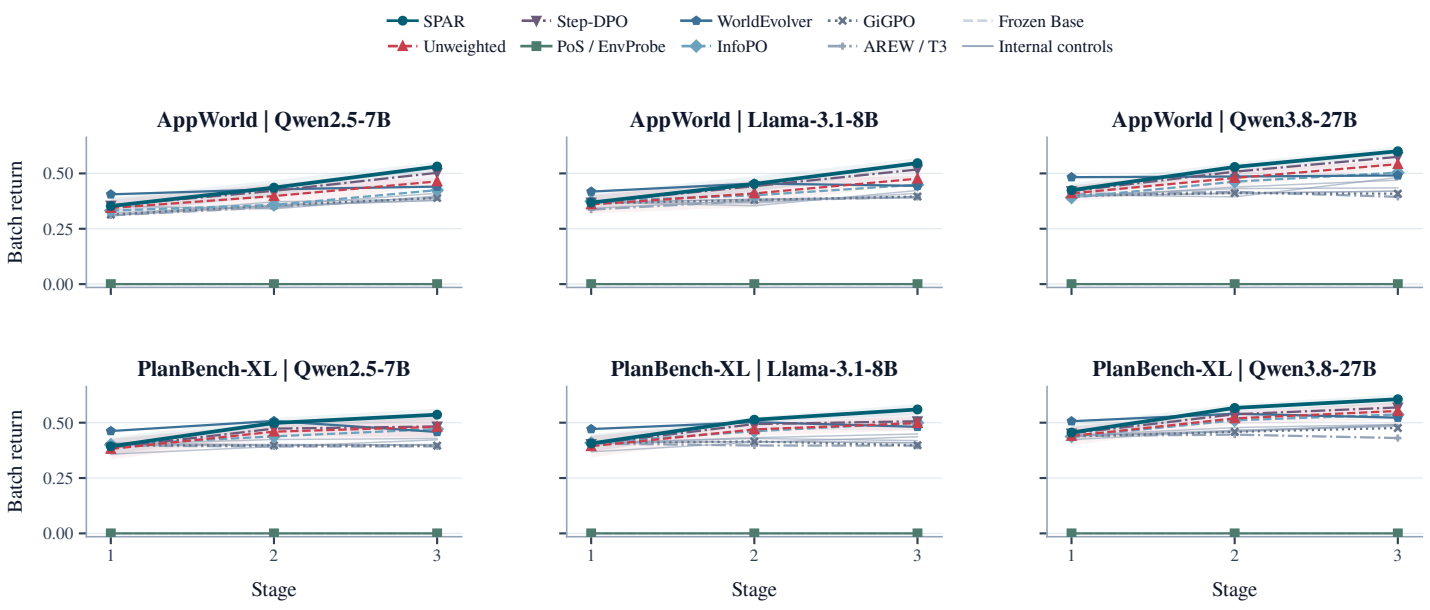

SPAR
Unweighted
Step-DPO
PoS / EnvProbe
InfoPO
WorldEvolver
GiGPO
AREW / T3
Frozen Base
Other controls
(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
AppWorld dev utility
0.8
0.6
0.4
0.2
0.6x
1.0x
1.4x
Budget Multiplier B/B_0

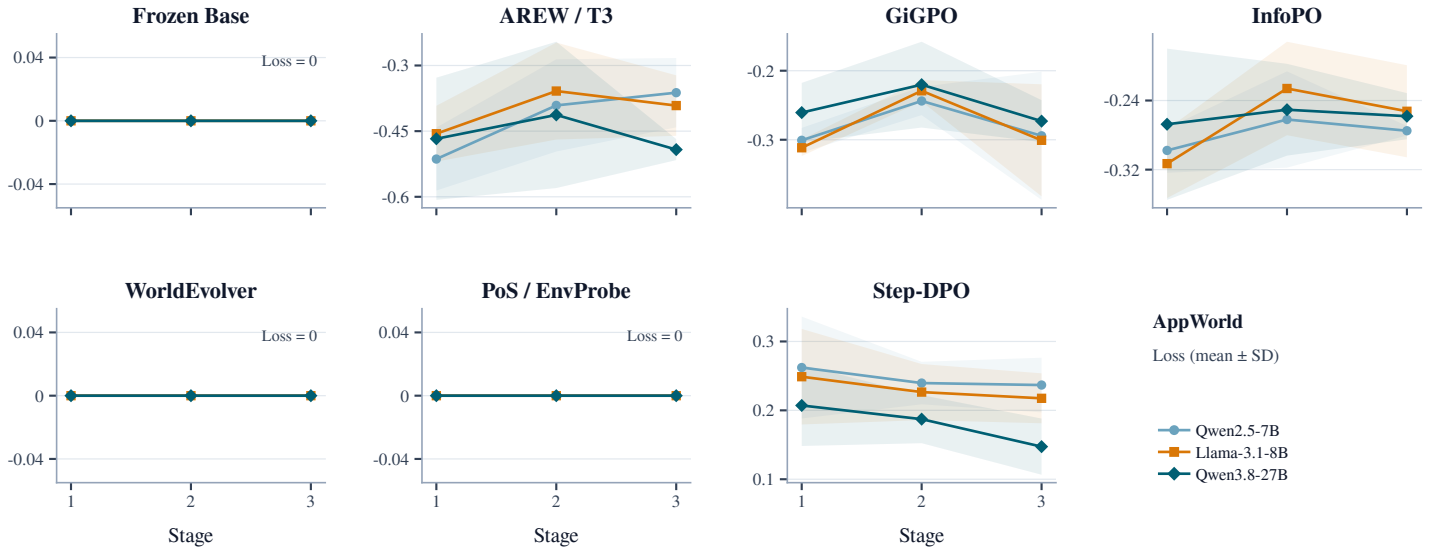

SPAR
Unweighted
Step-DPO
PoS / EnvProbe
InfoPO
WorldEvolver
GiGPO
AREW / T3
Frozen Base
Other controls
(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
AppWorld dev success (%)
100
75
50
25
0.6x
1.0x
1.4x
Budget Multiplier B/B_0

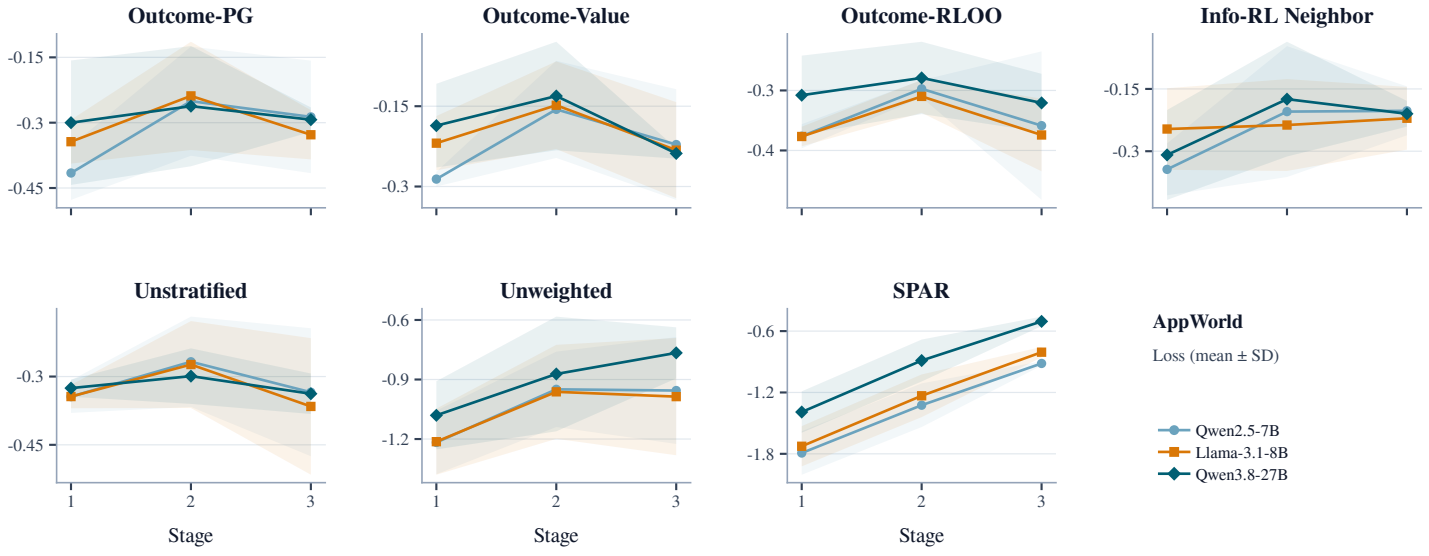
AppWorld dev

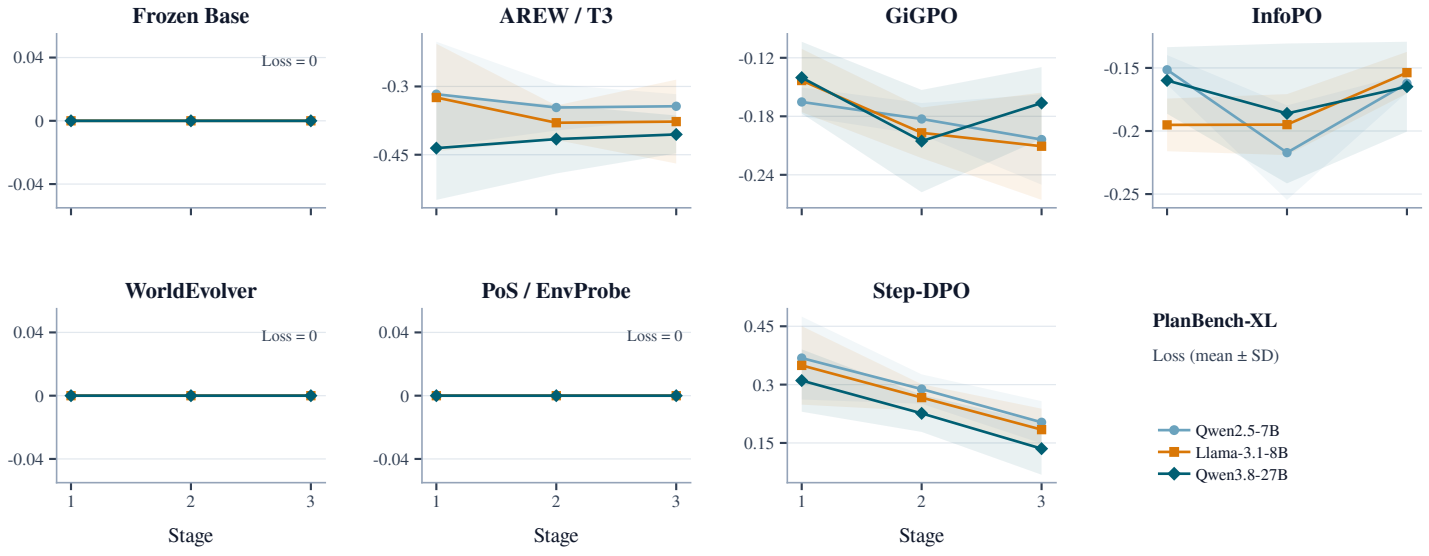
Method	Qwen2.5-7B			Llama-3.1-8B			Qwen3.8-27B		
	0.6x	1.0x	1.4x	0.6x	1.0x	1.4x	0.6x	1.0x	1.4x
Frozen Base	0.2043	0.2723	0.3014	0.2144	0.2877	0.3191	0.2567	0.3482	0.3874
AREW / T3	0.2166	0.2913	0.3234	0.2195	0.2954	0.3280	0.2910	0.3968	0.4422
GiGPO	0.2333	0.3154	0.3506	0.2567	0.3482	0.3874	0.3150	0.4323	0.4825
InfoPO	0.2844	0.3882	0.4327	0.3121	0.4282	0.4779	0.3674	0.5082	0.5685
WorldEvolver	0.3267	0.4487	0.5009	0.3326	0.4568	0.5101	0.3485	0.4805	0.5370
PoS / EnvProbe	0.3478	0.4800	0.5367	0.3602	0.5062	0.5688	0.4363	0.6430	0.7317
Step-DPO	0.3702	0.5145	0.5764	0.3891	0.5422	0.6079	0.4686	0.6554	0.7355
Outcome-PG	0.2275	0.3073	0.3414	0.2304	0.3113	0.3460	0.3158	0.4327	0.4829
Outcome-Value	0.2363	0.3195	0.3552	0.2596	0.3523	0.3920	0.3209	0.4405	0.4917
Outcome-RLOO	0.2275	0.3073	0.3414	0.2421	0.3277	0.3644	0.3129	0.4287	0.4783
Info-RL Neighbor	0.2596	0.3523	0.3920	0.2903	0.3964	0.4419	0.3405	0.4687	0.5236
Unstratified	0.2676	0.3641	0.4054	0.2982	0.4082	0.4553	0.3464	0.4768	0.5327
Unweighted	0.3883	0.5895	0.6750	0.3678	0.5374	0.6094	0.4199	0.6257	0.7166
SPAR	**0.4136**	**0.5967**	**0.6752**	**0.4290**	**0.6130**	**0.6918**	**0.4891**	**0.6906**	**0.7837**

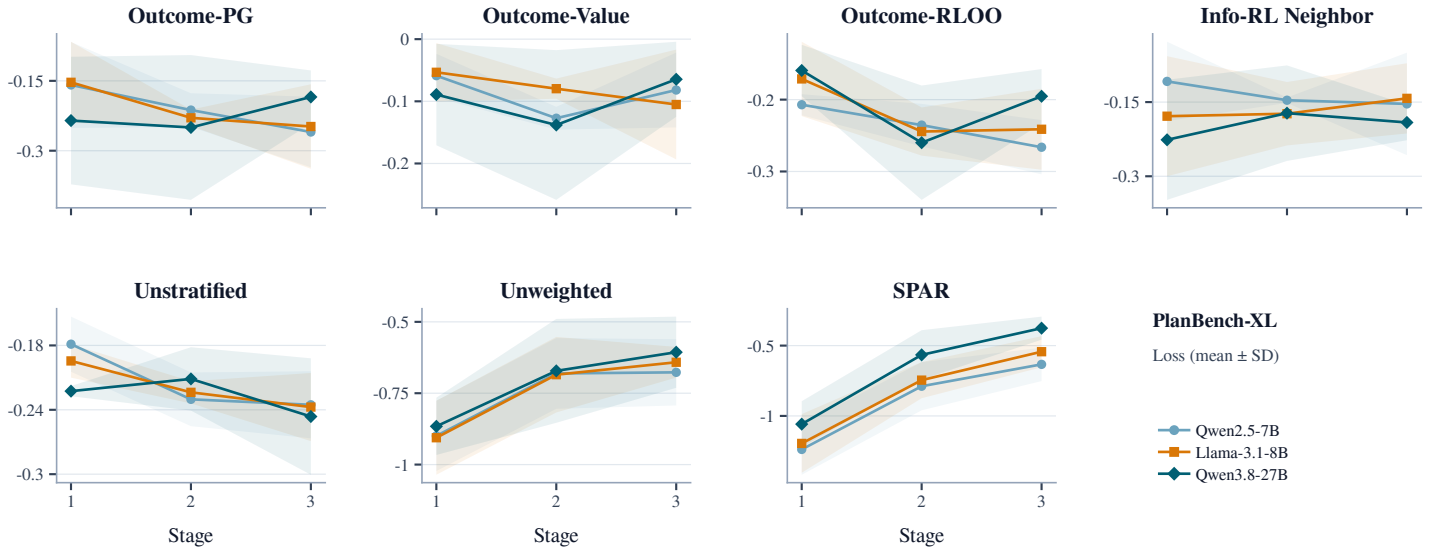

(a) Qwen2.5-7B
(b) Llama-3.1-8B
(c) Qwen3.8-27B
Test utility
Dev Δ_{use}
0.8
0.4
0.0
0.6130
0.4788
0.3758
0.6279
0.5052
0.4127
0.6948
0.6304
0.4908
0.5033
0.3612
0.3306
0.5579
0.3892
0.3513
0.6708
0.5208
0.4181
Root
$T = 0$
Uniform
$T \in \{0, 1\}$
Uniform
$T \in \{0, 1, 2\}$


SPAR
Unweighted
Step-DPO
PoS / EnvProbe
WorldEvolver
InfoPO
GiGPO
AREW / T3
Frozen Base
Internal controls
AppWorld | Qwen2.5-7B
AppWorld | Llama-3.1-8B
AppWorld | Qwen3.8-27B
PlanBench-XL | Qwen2.5-7B
PlanBench-XL | Llama-3.1-8B
PlanBench-XL | Qwen3.8-27B
Batch return
0.00
0.25
0.50
1
2
3
Stage


Frozen Base
Loss = 0
0.04
0
-0.04
AREW / T3
-0.3
-0.45
-0.6
GiGPO
-0.2
-0.3
InfoPO
-0.24
-0.32
WorldEvolver
Loss = 0
0.04
0
-0.04
PoS / EnvProbe
Loss = 0
0.04
0
-0.04
Step-DPO
0.3
0.2
0.1
1
2
3
Stage
AppWorld
Loss (mean ± SD)
Qwen2.5-7B
Llama-3.1-8B
Qwen3.8-27B

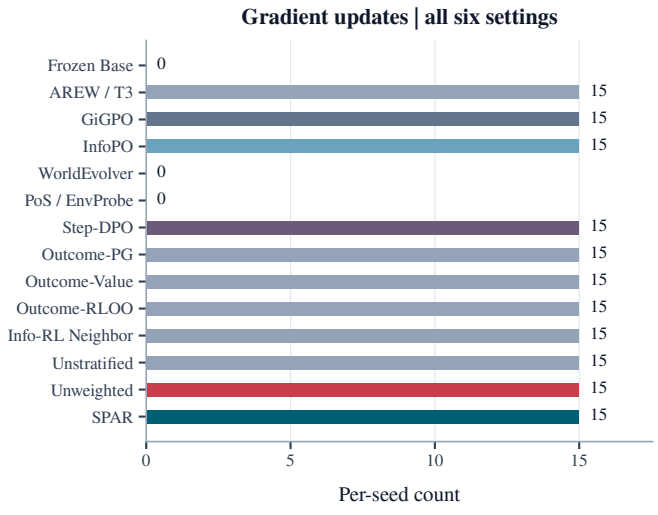

Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
SPAR
-0.15
-0.3
-0.45
-0.6
-0.9
-1.2
-1.8
1
2
3
Stage
AppWorld
Loss (mean ± SD)
Qwen2.5-7B
Llama-3.1-8B
Qwen3.8-27B

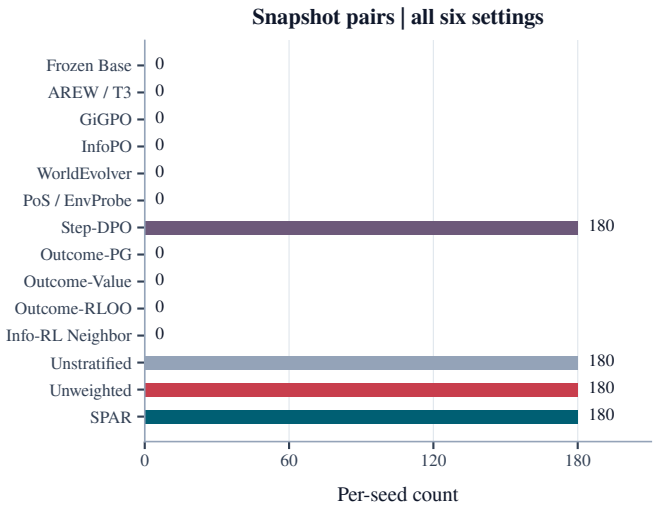

Frozen Base
Loss = 0
0.04
0
-0.04
AREW / T3
-0.3
-0.45
GiGPO
-0.12
-0.18
-0.24
InfoPO
-0.15
-0.2
-0.25
WorldEvolver
Loss = 0
0.04
0
-0.04
1
2
3
Stage
PoS / EnvProbe
Loss = 0
0.04
0
-0.04
1
2
3
Stage
Step-DPO
0.45
0.3
0.15
1
2
3
Stage
PlanBench-XL
Loss (mean ± SD)
Qwen2.5-7B
Llama-3.1-8B
Qwen3.8-27B


Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
SPAR
PlanBench-XL
Loss (mean ± SD)
Qwen2.5-7B
Llama-3.1-8B
Qwen3.8-27B
Stage
Stage
Stage

Environment steps / accounted tokens (k), mean per seed

	AppWorld			PlanBench-XL		
Method	Qwen2.5-7B	Llama-3.1-8B	Qwen3.8-27B	Qwen2.5-7B	Llama-3.1-8B	Qwen3.8-27B
Frozen Base	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0
AREW / T3	1434.0 / 489.0	1407.7 / 471.2	1405.7 / 479.7	1289.7 / 582.5	1307.0 / 570.3	1280.0 / 579.1
GiGPO	1428.7 / 487.1	1401.3 / 468.2	1397.7 / 475.3	1294.7 / 586.1	1299.7 / 566.4	1276.0 / 576.4
InfoPO	1426.0 / 488.0	1408.0 / 471.4	1371.0 / 460.9	1288.0 / 581.6	1269.3 / 545.9	1265.7 / 571.0
WorldEvolver	1319.7 / 582.9	1309.3 / 573.1	1306.3 / 579.8	1196.3 / 655.2	1193.3 / 632.5	1204.0 / 663.7
PoS / EnvProbe	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0	0.0 / 0.0
Step-DPO	1362.0 / 448.4	1365.0 / 445.1	1366.3 / 453.9	1303.3 / 593.7	1305.0 / 571.3	1309.7 / 603.2
Outcome-PG	1438.0 / 491.0	1402.7 / 469.2	1407.7 / 480.9	1306.3 / 594.5	1302.3 / 568.4	1271.0 / 572.8
Outcome-Value	1434.3 / 490.1	1396.7 / 466.0	1387.0 / 469.9	1298.3 / 589.8	1301.7 / 567.8	1269.7 / 572.2
Outcome-RLOO	1433.3 / 488.6	1397.3 / 465.0	1406.7 / 480.6	1304.3 / 593.3	1303.3 / 568.6	1271.0 / 572.8
Info-RL Neighbor	1436.3 / 493.5	1417.0 / 476.1	1392.3 / 471.8	1297.3 / 589.3	1287.3 / 558.1	1268.3 / 571.9
Unstratified	1435.7 / 489.9	1422.3 / 476.8	1381.0 / 462.1	1292.3 / 584.5	1292.3 / 561.1	1274.3 / 578.2
Unweighted	1372.0 / 454.0	1369.3 / 446.4	1368.7 / 454.5	1313.7 / 601.2	1314.0 / 578.0	1330.3 / 619.3
SPAR	**1367.3 / 450.5**	**1360.7 / 441.5**	**1366.7 / 453.6**	**1322.3 / 607.3**	**1323.7 / 584.5**	**1329.0 / 618.3**


Gradient updates | all six settings
Frozen Base
AREW / T3
GiGPO
InfoPO
WorldEvolver
PoS / EnvProbe
Step-DPO
Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
SPAR
0
15
15
15
0
0
15
15
15
15
15
15
15
15
0
5
10
15
Per-seed count


Snapshot pairs | all six settings
Frozen Base
AREW / T3
GiGPO
InfoPO
WorldEvolver
PoS / EnvProbe
Step-DPO
Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
SPAR
0
0
0
0
0
0
180
0
0
0
0
180
180
180
0
60
120
180
Per-seed count

Amortized model time / gradient-update time (s), mean per seed

	AppWorld			PlanBench-XL		
Method	Qwen2.5-7B	Llama-3.1-8B	Qwen3.8-27B	Qwen2.5-7B	Llama-3.1-8B	Qwen3.8-27B
Frozen Base	0.0 / 0.00	0.0 / 0.00	0.0 / 0.00	0.0 / 0.00	0.0 / 0.00	0.0 / 0.00
AREW / T3	150.2 / 26.84	89.4 / 26.54	277.1 / 4.40	82.7 / 2.58	113.6 / 2.76	322.4 / 2.46
GiGPO	149.0 / 27.66	89.0 / 65.24	274.2 / 4.41	83.2 / 2.56	112.5 / 2.70	320.5 / 2.48
InfoPO	147.9 / 29.41	90.6 / 56.07	265.7 / 5.23	82.9 / 3.29	109.5 / 3.47	317.0 / 3.15
WorldEvolver	124.3 / 0.00	81.5 / 0.00	245.4 / 0.00	75.0 / 0.00	99.5 / 0.00	289.3 / 0.00
PoS / EnvProbe	0.0 / 0.00	0.0 / 0.00	0.0 / 0.00	0.0 / 0.00	0.0 / 0.00	0.0 / 0.00
Step-DPO	124.4 / 35.91	85.1 / 21.75	261.1 / 6.17	85.1 / 4.35	115.7 / 4.63	337.4 / 4.24
Outcome-PG	152.0 / 22.52	89.0 / 35.16	277.9 / 5.26	84.4 / 3.29	112.9 / 3.55	318.2 / 3.22
Outcome-Value	150.3 / 31.89	88.6 / 69.34	271.0 / 3.81	83.9 / 2.72	113.2 / 2.88	317.8 / 2.54
Outcome-RLOO	150.5 / 31.49	88.4 / 35.66	277.8 / 3.81	84.2 / 3.28	112.9 / 3.51	318.3 / 3.12
Info-RL Neighbor	150.8 / 26.66	90.4 / 41.48	272.3 / 3.70	83.9 / 3.41	111.2 / 3.64	318.2 / 3.10
Unstratified	149.6 / 11.00	90.1 / 25.25	266.9 / 2.52	83.0 / 2.30	113.4 / 2.23	322.0 / 2.07
Unweighted	125.8 / 2.39	85.5 / 2.50	261.8 / 2.37	86.3 / 2.21	117.2 / 2.20	347.1 / 2.18
SPAR	**125.1 / 24.68**	**84.5 / 54.84**	**261.1 / 4.14**	**87.1 / 2.24**	**118.6 / 2.36**	**346.5 / 2.17**

Wall-clock time / snapshot-restore time (s), mean per seed

	AppWorld			PlanBench-XL		
Method	Qwen2.5-7B	Llama-3.1-8B	Qwen3.8-27B	Qwen2.5-7B	Llama-3.1-8B	Qwen3.8-27B
Frozen Base	0 / 0	<0.01 / 0	0 / 0	0 / 0	0 / 0	0 / 0
AREW / T3	30.65 / 0	30.52 / 0	7.88 / 0	5.73 / 0	6.12 / 0	5.42 / 0
GiGPO	31.51 / 0	68.98 / 0	7.92 / 0	5.70 / 0	5.96 / 0	5.51 / 0
InfoPO	33.26 / 0	60.01 / 0	8.95 / 0	6.49 / 0	6.96 / 0	6.31 / 0
WorldEvolver	3.93 / 0	3.98 / 0	3.63 / 0	3.49 / 0	3.73 / 0	3.45 / 0
PoS / EnvProbe	0.59 / 0	0.65 / 0	0.60 / 0	0.54 / 0	0.68 / 0	0.59 / 0
Step-DPO	40.13 / 0.007	26.03 / 0.003	10.12 / 0.010	7.97 / 0	8.52 / 0.003	7.85 / 0
Outcome-PG	26.32 / 0	38.96 / 0	8.79 / 0	6.38 / 0	6.98 / 0	6.25 / 0
Outcome-Value	35.87 / 0	73.27 / 0	7.38 / 0	5.85 / 0	6.35 / 0	5.48 / 0
Outcome-RLOO	35.42 / 0	39.54 / 0	7.48 / 0	6.41 / 0	7.00 / 0	6.07 / 0
Info-RL Neighbor	30.50 / 0	45.34 / 0	7.30 / 0	6.60 / 0	7.14 / 0	6.08 / 0
Unstratified	15.04 / 0.007	29.46 / 0	6.45 / 0.003	5.89 / 0.003	5.84 / 0	5.43 / 0
Unweighted	6.50 / 0.003	6.85 / 0.007	6.54 / 0.003	6.01 / 0.007	6.01 / 0.003	5.86 / 0.003
SPAR	**28.81 / 0**	**59.16 / 0.003**	**8.20 / 0.003**	**6.12 / 0**	**6.44 / 0**	**6.04 / 0.010**